

ABHINAV KAUSHIK

Senior AI Engineer

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SUMMARY

Senior AI/ML Engineer with 8+ years of experience delivering production AI and machine learning solutions across Healthcare, Telecom, AIOps, and client-facing technology environments. Experienced across the full lifecycle, from rapid prototypes and proof-of-concepts to production deployment, with expertise in GenAI, RAG, LangGraph, NLP, predictive analytics, anomaly detection, time-series forecasting, real-time ML systems, and automated evaluation and observability for AI agents. Strong experience translating business and client requirements into practical AI solutions and collaborating with technical and cross-functional teams. Currently building ThinkTree.AI, a non-profit AI-powered learning platform designed to support students with ADHD and context-switching challenges. Looking for opportunities in Berlin and willing to relocate.

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Optum

Bengaluru, India | Nov 2025 - Present

- Built and integrated a lab extraction agent with an existing Laboratory Information Management System (LIMS) using Claude Opus 4.7 and Amazon Textract, automating work associated with 800 Jira tickets and PRD tasks.
- Engineered a LangGraph-based analytics system for AARP medical plans, combining stateful multi-agent reasoning with real-time chart generation and reducing manual analysis effort by 98%.
- Built automated evaluation and observability pipelines for Question Answering and Summary Agents, leveraging LLM evaluators, quality metrics, tracing, and failure detection to reduce human-in-the-loop review.
- Enhanced dense retrieval performance by training and evaluating a cross-encoder with positive and negative examples for AARP survey data, achieving a 20% increase in retrieval accuracy.
- Implemented an automated evaluation framework for QA and summary agents using LLM evaluators and tracing, delivering scalable monitoring and reducing manual review effort.
- Enhanced dense retrieval by training a cross-encoder on positive and negative survey examples, integrating it into a vector-DB pipeline that improved relevance for content search.

Senior Machine Learning Engineer | Viavi Solutions

Chennai, India | Sep 2024 - Nov 2025

- Built 30+ AI application prototypes and mockups for clients across AIOps and GeoPerformance, translating client requirements into functional POCs for solution validation and demonstrations.
- Developed an AI-powered RAG-based Project Discovery Assistant using LangGraph, combining semantic retrieval and LLM reasoning to achieve a 90% match rate with existing organizational solutions.
- Deployed a production-grade SLA breach prediction system using Random Forest and LightGBM with FastAPI and MLflow, integrating GPT-4-powered reasoning for real-time root-cause analysis and remediation guidance, resulting in a 67% reduction in downtime.
- Built a LangGraph-driven RAG Project Discovery Assistant that retrieved semantic matches and generated recommendations, enabling efficient prototype validation for client solutions.

Data Scientist | Viavi Solutions

Bengaluru, India | Nov 2021 - Sep 2024

- Applied Sentence-BERT and DBSCAN to enhance semantic understanding and cluster alarm messages across graph-based AIOps infrastructure, achieving a 73.5% improvement in semantic alarm classification.
- Applied Isolation Forest and Holt-Winters for anomaly detection and demand forecasting across 500+ telecommunications network devices using 15+ KPIs, supporting earlier identification of abnormal time-series behavior and proactive network planning.
- Delivered technical POCs and consulting sessions on data services and products for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | United Health Group

Gurugram, India | Jun 2021 - Nov 2021

- Collaborated with stakeholders on ad-hoc analysis and customer-retention dashboards using Hive, SQL, and Power BI.
- Developed a Python- and Flask-based parser to automate data extraction, parsing, and transformation into structured data, supporting downstream analytics and reporting.

Data Engineer | Nippon Data Systems

Gurugram, India | Nov 2019 - May 2021

- Engineered SQL and PySpark-based data pipelines to automate BAU medical-audit reporting, transforming and aggregating large healthcare datasets and reducing manual reporting effort by 95%.
- Built real-time observability using Elasticsearch for centralized log ingestion and analysis, with Kibana dashboards for monitoring system activity and user events, improving traceability efficiency by 80% for the COVID QRT team.
- Developed a Flask-based enterprise search engine using Apache Solr and Lucene indexing, enabling centralized search and retrieval across organizational data for enterprise-wide use cases.

Associate Consultant | Saksoft

Gurugram, India | Dec 2017 - Nov 2019

- Conceptualized customer dashboards using QlikSense, SQL, and SAP HANA for financial and product insights, resulting in a 40% increase in user engagement.

- Developed an XGBoost-based propensity model to predict the probability of customers initiating a personal loan application within 7 days of contact, enabling targeted outreach and customer prioritization.
- Leveraged Selenium and Python to automate performance testing and evaluate application response times, resulting in a 90% reduction in testing effort.

PROJECTS

Think Tree | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Scalable Audio Content Understanding & Recommendation Engine

- Built an end-to-end pipeline that transcribed audio titles, generated chapter summaries with Claude, embedded segments using fine-tuned SBERT, and indexed them in pgvector.
- Evaluated relevance with cross-encoder re-ranking trained on click-through data, deployed the service via FastAPI with MLflow tracking, and monitored performance using Elasticsearch and Kibana.
- **Technologies:** Claude, SBERT, pgvector, LangGraph, FastAPI, MLflow, Elasticsearch, Kibana

Financial Fraud Detection | Mar 2025

- Developed an AI-driven fraud detection and reasoning workflow using the JPMorgan Synthetic Dataset, combining LightGBM-based modeling with LLM explanations to support interpretable financial-risk analysis.

Restaurant Feedback Analyzer | Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP automation platform using an LLM-driven supervisor to orchestrate intent detection, retrieval, proposal generation, document matching, and market research with RAG over historical proposals.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 - Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 - Aug 2017

TECHNICAL SKILLS

AI / GenAI: GenAI, LLM, RAG, Agentic AI, LangChain, LangGraph, Claude, OpenAI, Embeddings, Tool Orchestration, Cross-Encoder, Transformers, NLP, Generative AI, GenAI-based product features, innovation

Machine Learning: XGBoost, LightGBM, Random Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning, methodologies, experimental

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, Flask, Hive, SparkQL, REST API, Pydantic, Unix

MLOps / Cloud: MLflow, Docker, Kubernetes, Jenkins, Bamboo, Bitbucket, DVC, Terraform, Airflow, GCP, Azure, NCI Servers, Large-Scale Distributed Computation, productionize

Search / Observability: Elasticsearch, Kibana, Logging, Log Analysis, Apache Solr, Lucene Indexing, Search and Information Retrieval, AI Evaluation, LLM Evaluators, Agent Tracing

Data Platforms: Databricks, Apache Spark, Airflow

Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

BI / Testing: Power BI, QlikSense, SAP HANA, Selenium

LANGUAGES

English: Native
German: A1, Pursuing

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies